

①

Practice Problem

int a=0 b=0; $\rightarrow O(1)$
 for (i=0; i < N; i++) $\rightarrow \frac{O(1)}{0 \rightarrow n-1} \rightarrow O(n)$

{ for (j=0; j < N; j++) $O(1) \rightarrow O(n)$

{ a = a + j; $\rightarrow O(n(n))$

}
 for (k=0; k < N; k++) $\rightarrow O(n)$

b = b + k; $\rightarrow O(1)$

$$1 + n + n^2 + n$$

$$= O(n^2)$$

for (i=0; i < n; i++) $\rightarrow n+1$

{

} stmt; $\rightarrow n$

$$O(n)$$

or

$$\Rightarrow 1 + 1 + 1 + 1 + 1 + \dots + 1 = n$$

$$[k=n]$$

② ~~int~~ int a = 0

for (i = 0 ; i < N ; i++)

{

for (j = N ; j > i ; j--)

{

a = a + i + j

}

}

let N = 5

i	j	Count
0	5 to 1	5
1	5 to 2	4
2	5 to 3	3
3	5 to 4	2
4	5 to 5	1
5	0	0

$$N_2 5 = 5 + 4 + 3 + 2 + 1$$

$$N_2 N = N + N-1 + N-2 + N-3 + \dots + N-N$$

$$n + (n-1) + (n-2) + \dots + 1$$

$$= \frac{n(n+1)}{2}$$

$$= \frac{n^2}{2} + n$$

$$= \boxed{O(n)^2}$$

③

for (x=1; x<=n; x++)

for (y=1; y<=n; y++)

for (z=1; z<=n; z++)

$$= \boxed{O(n^3)}$$

④

if (x < z)

print("x is less than z");

else

{ for (i=1; i<=n; i++)
print i;

}

$$\boxed{O(n)}$$

⑤ for ($x=1$; $x < n$; $x *= 2$)
 print x ;

if $x = 1$

$n = 10$

$x = 1$

1

2

1

4

1

8

1

$n = 10$

1 + 2 + 4 + 8 + (4)

$n = 16$

1 + 2 + 4 + 8 + 16

$$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + \dots$$

$i=0 \quad i=1 \quad i=2 \quad i=3 \quad i=4$

$$= \log(n)$$

$$\leftarrow \boxed{2^k - n}$$

for ($i = n$; $i > 0$; $i = i/2$)
 {
 stmt
 }

$n+1$

3

$$\boxed{O(n)}$$

Ques for ($i = 1 ; i \leq n ; i = i + 2$)
{
 stmt;
}

—————→ $\frac{n}{2}$

$$f(n) = \frac{n}{2}$$

Ques for ($i = 1 ; i \leq n ; i = i + 10$)
{
 stmt;
}

—————→ $\frac{n}{10}$

$$f(n) = \frac{n}{10}$$

Ques for ($i = 1 ; i \leq n ; i++$) → $n+1$

for ($j = 1 ; j \leq n ; j++$) → $n(n+1)$

{
 Statement;
}

$$f(n) = \frac{n^2}{2}$$

```
for (i = 0; i < n; i++)
{
    for (j = 0; j < i; j++)
    {

```

stmt;

}

3

i =

j

no of times f(n)

0

0 x

0

1

0 ✓
1 x

1

2

0 ✓
1 ✓
2 x

2

3

0 ✓
1 ✓
2 ✓
3 x

3

⋮

⋮

n

⋮

(n-1)

$$= 0 + 1 + 2 + 3 + 4 + \dots + n$$

$$= \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2} = O(n^2)$$

Ques
Ans
 $P = 0$

for ($i = 1, P \leq n, i++$)

{

$P = P + i;$

}

i

P

$P(n)$ time

0/1

$0+1$

2

$2+1=3$

3

$= 1+2+3$

4

$1+2+3+4$

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

Assume. $P > n$

$P = \frac{K(K+1)}{2} > n$

2

$\frac{K^2 + K}{2} > n$

$> n$

~~$\frac{K^2 + K}{2} > n$~~

$O(\sqrt{n})$

Ques

for ($i = 1; i < n; i = i * 2$)

{

statement;

i

f

$1 + 2 + 4 + 8 + \dots + 2^k$
 $2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^k$

2

~~$2^{k+1} - 2$~~

$$i = 2^k$$

$$2^k \geq n$$

$$k = (\log_2 n)$$

Ans

Math.

for ($i = 1 ; i < n ; i = i * 2$)

$$i = 1 \times 2 \times 2 \times 2 \times 2 = n$$

$$2^k = n$$

$$k = \log_2 n$$

Ques ~~for~~ $(i = n$

$\text{for } (i = n, i > 1; i = i/2)$

{

Statement

} Statement count

$\frac{n}{1} - 1$

$\frac{n}{2} - 1$

$\frac{n}{4} - 1$

$\frac{n}{8} - 1$

$\frac{n}{2^k} - 1$

assume $i < 1$

$\frac{n}{2^k} < 1$

$= n < 2^k$

$k = \log_2 n$

$O(\log_2 n)$

Ques

$\text{for } (i = 0; i * i < n; i++)$

{ Statement

terminate condition $i * i \geq n$

$i^2 = n$

$i = \sqrt{n}$

Ques

$P=0$

for ($i=1 ; i < n ; i = i * 2$)
 $\left[\begin{array}{l} \{ \\ P++ ; \\ \} \end{array} \right] \log(n)$

for ($j=1 ; j < P ; j = j * 2$)
 $\left[\begin{array}{l} \{ \\ \text{statement} \\ \} \end{array} \right] \log(P)$

$1 \times 2 \times 2 \times \dots \times 2$
 $2^k = P$
 $\log P = k$

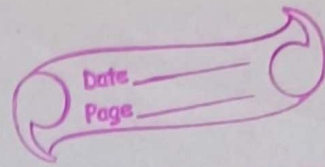
$\log \cdot \log(n)$

Ans

for ($i=0 ; i < n ; i++$)
 $\{$
 for ($j=1 ; j < n ; j = j * 2$) $\log n$
 $\left[\begin{array}{l} \{ \\ \text{statement} \\ \} \end{array} \right] n \times \log n$
 $\}$

$O(n \log n)$

Note



for ($i=0, i < n, i++$) $O(n)$

for ($i=0, i < n, i+=i+2$) $O(n/2)$

for ($i=n, i > 1, i--$) $O(n)$

for ($i=1, i < n, i=i*2$) $O(\log_2 n)$

for ($i=1, i < n, i=i*3$) $O(\log_3 n)$

for ($i=n, i > 1, i=i/2$) $O(\log_2 n)$